Using Chemical Modeling to Understand the Impact of Temperature on Lithium-Ion Battery Performance

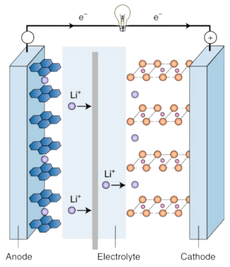
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INTRODUCTION

Lithium Ion Batteries:

- created in the 1970s with the goal of replacing fossil fuels
- used to power modern devices, such as electric cars and cell phones
- recharged through the movement of chemical electrodede between the anode and cathode ends of the battery



Temperature and Surroundings:

- environmental factors play a crucial role in the performance of a battery
- drastic temperature changes or extreme conditions occassionally lead to fluctuations in battery performance

LITERATURE REVIEW

2018 **A Review of Lithium-ion Battery Safety Concerns: The Issues, Strategies, and Testing Standard**

- examines the effects of thermal energy on lithium-ion batteries' safety and performance

2020 **Predictive modeling of battery degradation and greenhouse gas emissions from U.S. state-level electric vehicle operation**

- explains how to develop predictive models for battery degradation and performance

RESEARCH OBJECTIVES

Optimization:
Especially as technology is advancing, understanding the limitations of Li-ion batteries is an extremely crucial aspect of battery design.

Using chemical modeling and data analysis methodologies, our research will:

1. Determine a relationship between ambient temperature and battery capacity+internal resistance
2. Find the optimal temperature range in which Li-ion batteries will perform the best

METHODOLOGY

Performance Metrics:
The performance of the Li-ion battery is determined by **battery capacity** and **internal resistance**.

- **Battery Capacity (ampere-hours):** amount of energy a given battery can store, directly proportional to battery performance
- **Internal Resistance (Ohms):** the opposition of the flow of a current in the battery, indirectly proportional to battery performance

PyBaMM:
Python package that can chemically model Li-ion batteries and simulate chemical reactions.

MODEL
We used the Single-Particle-Model Lithium ion battery model with standard Chen2020 parameters.

SIMULATION
We altered the Ambient Temperature parameter and ran the simulation for 3600 seconds.

PyBaMM

Data Collection:

- We simulated temperatures from 250 Kelvin to 350 Kelvin with increments of 10 Kelvin, collecting the data on the final Capacity and Internal Resistance value after 1 hour.
- These temperature ranges were chosen because they best-matched household temperature extremes.
- For the environment of the simulation, it is better to use the virtual lithium-ion battery and computational methods to simulate these conditions because there would be more challenges associated with physical investigation

Data Visualization and Analysis:

- Using MatPlotLib.pyplot, we graphed our data to get Battery Capacity vs. Temperature and Internal Resistance vs. Temperature graphs.
- Each graph will have temperature in Kelvin as the independent variable on the x-axis. Capacity (Ah) and internal resistance (ohms) are measured on the y-axis in their respective graphs.

Data Validation and Interpretation:

- The data was validated against known theoretical models and existing experimental findings in the literature.
- The validated results were then interpreted in the context of practical applications, providing guidance for battery design and usage in environments where temperature is a critical factor.

RESULTS

	Temperature	Capacity	Internal Resistance
0	250	0.3200094	0.737535
1	260	0.523247	0.807640
2	270	0.662300	0.867636
3	280	0.673295	0.917998
4	290	0.680616	0.956067
5	300	0.680616	0.848601
6	310	0.680616	0.779865
7	320	0.680616	0.730946
8	330	0.680616	0.694632
9	340	0.680616	0.668237
10	350	0.680616	0.649742

TABLE 1. SIMULATION RESULTS: EFFECT OF TEMPERATURE ON BATTERY CAPACITY AND INTERNAL RESISTANCE

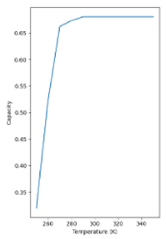


FIGURE 1. THE VARIATION IN BATTERY CAPACITY ACROSS THE SIMULATED TEMPERATURE RANGE OF 250K TO 350K.

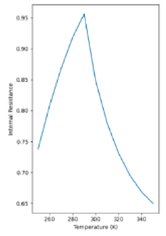


FIGURE 2. THE VARIATION IN INTERNAL RESISTANCE ACROSS THE SIMULATED TEMPERATURE RANGE OF 250K TO 350K.

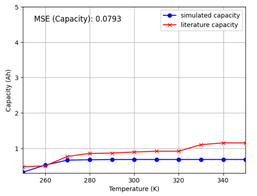


FIGURE 3. COMPARISON OF SIMULATED AND LITERATURE CAPACITY VS. TEMPERATURE

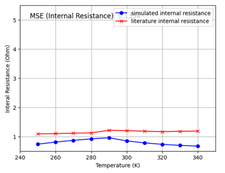


FIGURE 4. COMPARISON OF SIMULATED AND LITERATURE INTERNAL RESISTANCE VS. TEMPERATURE

DISCUSSION

Battery Capacity:
Similarly, **battery capacity increased with temperature and stabilized around 280 K**. This results from increased temperature leading to increased ion mobility.

Internal Resistance:
Looking at the graphs, **internal resistance increases with temperature and peaks at 290 Kelvin**. However, after temperature exceeds 290 Kelvin, internal resistance decreases. This is most likely caused by increased ionic conductivity within the battery's chemical components.

MSE Values:
Using existing data from our literature review, the calculated MSE value for our battery capacity data was smaller than the MSE value for our internal resistance data.

CONCLUSION

Relationships:

- Battery Capacity: increases as temperature increases but eventually plateaus at a maximum of 0.68 Ah at 280 K.
- Internal Resistance: increases as temperature increases with a peak of 0.87 Ohms at 290 K before decreasing when the temperature exceeds the maximum.

Optimal Temperature Range:

290 K - 300 K — Decreased internal resistance and maximum battery capacity

ACKNOWLEDGMENTS & REFERENCES

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